

The English lab

For the English language lab, we have designed activities that students can perform repeatedly till they are thorough with a concept. In a real-world tutorial class, a teacher will need to review the work done by students regularly, and give feedback, so that they can improve their performance. This takes up too much of the teacher's time, and limits the extent of tutorial activities. In OLABs, reviewing the students' solutions is done by our system, automatically. The feedback is directly dependent on the selected problem and the mistakes made by the student at that particular time. This helps students to notice the problems in their solutions and improve on them with practice.

Another interesting aspect of the English lab is the use of templates to generate sentences, instead of using a set of canned sentences for the activities. This helps prevent blind copying from others, and offers a wide range of sentences for practice. Sophisticated language processing techniques are deployed to make this effective.

English labs are currently available for topics such as tense conversion, voice conversion, use of prepositions, subject-verb agreements, etc.

Current status

The system has been available on the OLABs website for the past few years. There are over 150 labs covering physics, chemistry, biology, mathematics and English. Twelve English labs are currently open to students of different classes, depending on the concepts that they want to practice. The number of other labs is given below (as per the CBSE curriculum).

Class/ subject	Physics	Chemistry	Biology	Math
IX	9	11	9	13
X	13	10	6	12
XI	16	10	13	-
XII	16	15	8	-

OLabs is gaining a lot of attention through various channels, with 50,000 to 100,000 page views a week, depending on the academic calendar. Since many schools do not have the required Internet access for regular use in classrooms or labs, a live DVD version of the entire OLABs system has been developed. There is also a Windows installer for schools to set up the system on their computers, locally.

In 2013, the then CBSE chairman, Dr Vineet Joshi, had written to all schools encouraging them to use OLABs, and recently, another circular has been issued with the same message. It also encourages CBSE teachers to benefit from the training that CDAC offers. In December 2015, the union minister for communications and IT, Honourable Shri Ravi Shankar Prasad, formally launched a programme

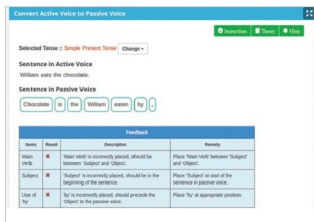


Figure 3: English lab on voice conversion

to train 30,000 teachers on OLABs across India, under the Digital India initiative. This training is free, and we invite teachers from CBSE schools to register on the portal for this programme. We will schedule these programmes in various parts of the country, and will inform those who register, on the participation dates. Over 5000 teachers have been trained so far.

The technologies used

The entire system is built on open source software, and also follows open standards, to ease localisation and other modifications. The technologies used are listed below.

1. Client side: HTML5, JavaScript CCS3, KineticJS, Three.js and JQuery
2. Server side: PHP, JSP and Servlets
3. Database: MySQL

The UI design is responsive; hence, we can use websites in the desktop environment as well as on handheld devices. The development follows software engineering principles, once again using open source software such as Git (code versioning), GitLab (issue tracking and repository management) and Eclipse (IDE) among others.

Acknowledgements

The success of the project owes much to the great support we received from DeitY, government of India, especially from Dr Ajay Kumar, Dipak Singh, D.K. Kalra, Santosh Pandey and A.K. Arora. I also thank the OLABs team at CDAC Mumbai, particularly the team leads Archana Sharma and Manoj Kumar Singh, and the team at Amrita Vishwa Vidyapeetham, headed by Prof. Prema N., for the passion and commitment they bring to the project. Special thanks to Archana for the critical review of this article. 

By: Dr M. Sasikumar

The author is associate director at CDAC Mumbai, and is the chief investigator of the project. He can be reached at sasi@cdac.in